# Capacity Utilisation

Exam code: 9BS0

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## Calculating capacity utilisation

- Capacity utilisation is a **measure of the level** to which a business's **assets** are being used to produce output
- It **compares the current output to the maximum possible output** a business can produce using all of its assets and is expressed as a percentage
- Capacity utilisation is calculated using the formula

$$\text{Capacity utilisation} = \frac{\text{Current output}}{\text{Maximum possible output}} \times 100$$

### Worked Example

Lola Bakery produces specialist Indian and Bangladeshi breads that are sold to restaurants in the West Midlands. Batch production is used in the factory to manufacture a range of breads, and the factory can produce a maximum of 68,400 units per month.

In May, factory output was 51,420 units.

Calculate Lola Bakery's capacity utilisation in May.

**Step 1: Divide the current output by the maximum output**

$$= \frac{51{,}420 \text{ units}}{68{,}400 \text{ units}}$$

$$= 0.75$$

**Step 2: Multiply the outcome by 100 to obtain the percentage capacity utilisation**

$$= 0.75 \times 100$$

$$= 75\%$$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it
Help me understand

### Worked Example

range of breads, and the factory can produce a maximum of 68,400 units per month.

In August, capacity utilisation increased to 92%.

Calculate the number of units produced by Lola Bakery in August.

**Step 1: Convert 92% to a decimal**

$$= 92\% \div 100$$

$$= 0.92$$

**Step 2: Find 92% of the maximum output using the decimal**

$$= 0.92 \times 68{,}400$$

$$= 62{,}928 \text{ units}$$

# Implications of underutilisation and overutilisation of capacity

## Underutilisation of capacity

- If a business has a low level of capacity utilisation, it will **not be making the most of its resources** and is likely to have **increased unit costs**
  - The **fixed costs** are spread over fewer units of output, resulting in higher average total costs
  - Workers may be **underdeployed**, leading to fears of **redundancy**
- Operating under capacity does provide a business with **flexibility**
  - There may be an opportunity to engage workers in **maintenance tasks**
  - The business can respond to **sudden increases in demand**

## Overutilisation of capacity

- If a business has a high level of capacity utilisation, it may not have the **flexibility to respond to new orders** from customers
  - Staff will be under a lot of pressure to produce high levels of output
  - Overworked staff may be inclined to leave, increasing **staff turnover**
  - Machinery may be pushed to its limits and prone to **breakdowns**, which **disrupts production and increases costs**
- High capacity utilisation will **minimise average total costs** and increase business **competitiveness**
  - If workers are busy, they are likely to **feel secure in their employment**

## Worked Example

**Production data for pencil manufacturers A and B**

| Manufacturer | Current output (units) | Maximum output (units) |
|---|---|---|
| **A** | 3m | 6m |
| **B** | 2.4m | 4m |

Explain **one** implication of the level of capacity utilisation for pencil manufacturer A, compared to manufacturer B. You are advised to show your workings.

[4]

**Step 1: Demonstrate knowledge of capacity utilisation**

- Capacity utilisation measures how well a business is using its available production capacity. **[1]**

**Step 2: Calculate the capacity utilisation for both manufacturers**

$$\text{Manufacturer A} = \frac{3 \text{ million units}}{6 \text{ million units}} \times 100 \quad \textbf{[1]}$$

$$= 50\%$$

$$\text{Manufacturer B} = \frac{2.4 \text{ million units}}{4 \text{ million units}} \times 100 \quad \textbf{[1]}$$

$$= 60\%$$

**Step 3: Explain the implications**

One implication is that staff may be bored and demotivated, particularly at manufacturer A, because they have less work to do than at manufacturer B. **[An]**

Or:

Unit costs are likely to be higher than they could be at both manufacturers **[K]** because resources such as workers and machinery are not being used to their full potential **[An]**.

## Examiner Tips and Tricks

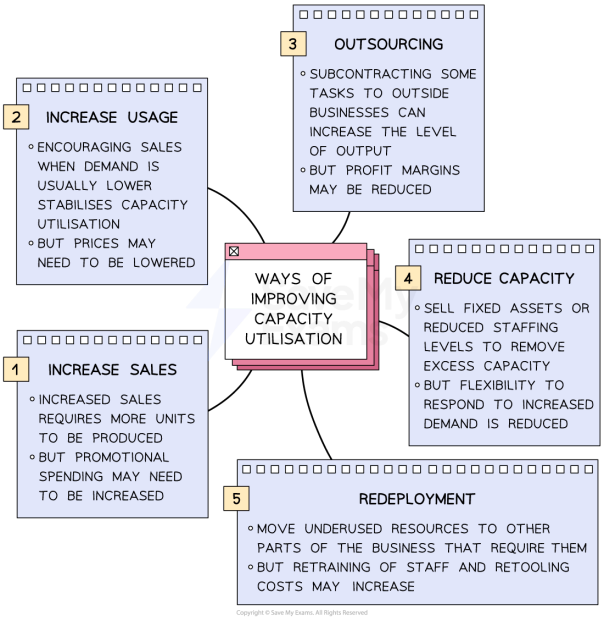
Including a definition of the key term is not essential, as **implicit understanding** of a business term or concept can be shown through correct calculations and identifying appropriate points.

In the example above, the definition could attract a mark if calculations had not been included, but the answer achieved

# Ways of improving capacity utilisation

- Achieving an **optimum** level of capacity utilisation is a key **production objective**
- A business has several options available to improve its **capacity utilisation**

## A range of ways to improve capacity utilisation




***Ways of improving capacity utilisation include increasing sales, reducing capacity and using outsourcing or redeployment***

- Increasing the level of capacity utilisation results in lower unit costs
- In some cases, **increasing demand and thus output** is the most suitable way to increase capacity utilisation
  - For example, reducing selling prices or offering short-term **price promotions** can increase demand
  - If **competitors go out of business**, demand should pick up, which will improve capacity utilisation without the need for further promotion

- **Reducing the level of overall capacity** will improve the level of capacity utilisation
  - Capacity can be reduced by selling capital equipment or reducing the number of staff

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